In [1]:
```
pip install nltk scikit-learn
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: nltk in c:\programdata\anaconda3\lib\site-packages (3.9.2)
Requirement already satisfied: scikit-learn in c:\programdata\anaconda3\lib\site-packages (1.
7.2)
Requirement already satisfied: click in c:\programdata\anaconda3\lib\site-packages (from nltk)
(8.2.1)
Requirement already satisfied: joblib in c:\programdata\anaconda3\lib\site-packages (from nlt
k) (1.5.2)
Requirement already satisfied: regex>=2021.8.3 in c:\programdata\anaconda3\lib\site-packages
(from nltk) (2025.9.1)
Requirement already satisfied: tqdm in c:\programdata\anaconda3\lib\site-packages (from nltk)
(4.67.1)
Requirement already satisfied: numpy>=1.22.0 in c:\programdata\anaconda3\lib\site-packages (fr
om scikit-learn) (2.3.5)
Requirement already satisfied: scipy>=1.8.0 in c:\programdata\anaconda3\lib\site-packages (fro
m scikit-learn) (1.16.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\programdata\anaconda3\lib\site-packa
ges (from scikit-learn) (3.5.0)
Requirement already satisfied: colorama in c:\programdata\anaconda3\lib\site-packages (from cl
ick->nltk) (0.4.6)
Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
from nltk import pos_tag
```

In [3]:
```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\Adil\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\Adil\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\Adil\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     C:\Users\Adil\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
```

Out[3]: True

In [4]:
```
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     C:\Users\Adil\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\Adil\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

Out[4]: True

In [6]:
```
text = "Data science is an interdisciplinary field that uses scientific methods, processes, a
print(text)
```

```
Data science is an interdisciplinary field that uses scientific methods, processes, algorithms
and systems to extract knowledge from data.
```

In [7]:
```
tokens = word_tokenize(text)
print(tokens)
```

```
['Data', 'science', 'is', 'an', 'interdisciplinary', 'field', 'that', 'uses', 'scientific', 'm
ethods', ',', 'processes', ',', 'algorithms', 'and', 'systems', 'to', 'extract', 'knowledge',
'from', 'data', '.']
```

In [8]:
```
pos_tags = pos_tag(tokens)
print(pos_tags)
```

```
[('Data', 'NNP'), ('science', 'NN'), ('is', 'VBZ'), ('an', 'DT'), ('interdisciplinary', 'JJ'),
('field', 'NN'), ('that', 'WDT'), ('uses', 'VBZ'), ('scientific', 'JJ'), ('methods', 'NNS'),
(',', ','), ('processes', 'NNS'), (',', ','), ('algorithms', 'NN'), ('and', 'CC'), ('systems',
'NNS'), ('to', 'TO'), ('extract', 'VB'), ('knowledge', 'NN'), ('from', 'IN'), ('data', 'NNS'),
('.', '.')]
```

In [9]:
```
stop_words = set(stopwords.words('english'))

filtered_words = [word for word in tokens if word.lower() not in stop_words]
print(filtered_words)
```

```
['Data', 'science', 'interdisciplinary', 'field', 'uses', 'scientific', 'methods', ',', 'proce
sses', ',', 'algorithms', 'systems', 'extract', 'knowledge', 'data', '.']
```

In [10]:
```
ps = PorterStemmer()

stemmed_words = [ps.stem(word) for word in filtered_words]
print(stemmed_words)
```

```
['data', 'scienc', 'interdisciplinari', 'field', 'use', 'scientif', 'method', ',', 'process',
',', 'algorithm', 'system', 'extract', 'knowledg', 'data', '.']
```

In [11]:
```
lemmatizer = WordNetLemmatizer()

lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_words]
print(lemmatized_words)
```

```
['Data', 'science', 'interdisciplinary', 'field', 'us', 'scientific', 'method', ',', 'proces
s', ',', 'algorithm', 'system', 'extract', 'knowledge', 'data', '.']
```

In [12]:
```
from sklearn.feature_extraction.text import TfidfVectorizer

documents = [text]

vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(documents)

print("Feature Names:", vectorizer.get_feature_names_out())
print("TF-IDF Values:\n", tfidf_matrix.toarray())
```

```
Feature Names: ['algorithms' 'an' 'and' 'data' 'extract' 'field' 'from'
 'interdisciplinary' 'is' 'knowledge' 'methods' 'processes' 'science'
 'scientific' 'systems' 'that' 'to' 'uses']
TF-IDF Values:
 [[0.21821789 0.21821789 0.21821789 0.43643578 0.21821789 0.21821789
  0.21821789 0.21821789 0.21821789 0.21821789 0.21821789 0.21821789
  0.21821789 0.21821789 0.21821789 0.21821789 0.21821789 0.21821789]]
```